
HorizonBench: A Comprehensive AI-Powered Academic Paper Review System

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Abstract

1 Introduction

1.1 Problem Statement

1.2 Project Goals

1.3 Significance

2 Literature Review

Automated evaluation of academic papers lies at the intersection of natural language processing, machine learning, and scholarly communication. This section surveys the evolution from traditional automated essay scoring (AES) to contemporary LLM-based approaches, examines multi-agent consensus mechanisms, and identifies research gaps motivating HorizonBench. Figure 1 provides a visual taxonomy of this evolution.

2.1 Automated Essay Scoring: From Feature Engineering to Deep Learning

The field of Automated Essay Scoring originated in the 1960s with Page’s Project Essay Grade (PEG), which employed surface-level features such as sentence length and lexical complexity [1]. While PEG demonstrated feasibility, its reliance on superficial proxies precluded evaluation of argumentative logic—limitations pronounced when assessing academic papers where methodological rigor constitutes primary value.

The Educational Testing Service’s e-rater advanced the field by integrating grammar detection and coherence features with machine learning [2]. However, dependence on manually crafted features limited generalization to academic-specific dimensions such as experimental design validity.

The deep learning era brought substantial improvements. Taghipour and Ng [3] introduced BiLSTM architectures for document-level semantic understanding. Dong et al. [4] combined CNNs with attention mechanisms for local pattern capture. Transformer-based approaches subsequently achieved state-of-the-art results: Wang et al. [5] proposed multi-scale BERT representations, while hybrid approaches combining RoBERTa with linguistic features demonstrated QWK scores exceeding 0.92 [6].

However, Misgna et al. [7] emphasize that feedback generation remains largely overlooked. Deep learning models capture general logical patterns but cannot interpret specialized technical logic—for instance, evaluating whether an ablation study sufficiently covers key variables.

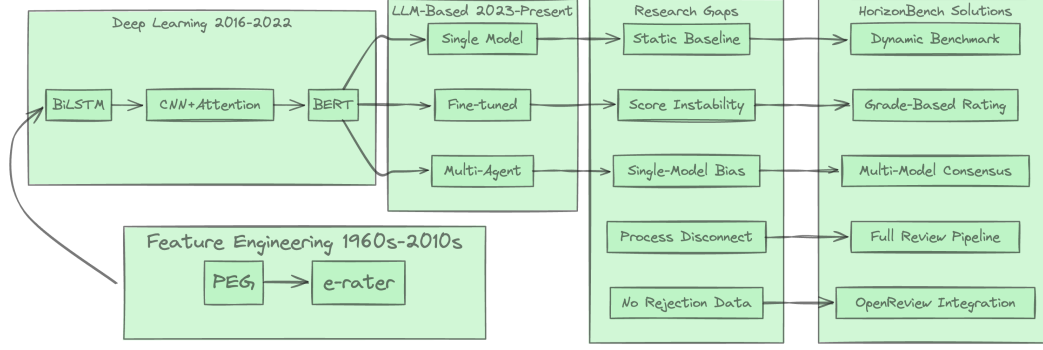


Figure 1: Taxonomy of Automated Academic Assessment: Technical Evolution and Research Gap Analysis. The diagram traces three eras of development and maps identified research gaps to HorizonBench solutions.

2.2 LLM-Based Automated Scholarly Paper Review

Large Language Models have catalyzed a paradigm shift toward Automated Scholarly Paper Review (ASPR). A comprehensive survey by Tan et al. [8] documents this transformation, noting that LLMs’ long-context capabilities enable simulation of complete review processes. Systems like ReviewMT [9] convert peer review into multi-turn dialogues with LLMs acting as authors, reviewers, and decision-makers.

Zhou et al. [10] evaluated LLM reliability using the RR-MCQ benchmark from ICLR-2023. Their findings reveal that while LLMs achieve passable accuracy (>60%) on single options, completely correct answers remain rare (20%), with particular weakness in long paper processing and critical feedback generation.

OpenReviewer [11] fine-tuned Llama on 79,000 reviews from top ML conferences, demonstrating that specialized training can overcome LLMs’ tendency toward overly positive assessments. The SEA framework [12] introduces cross-venue standardization with self-correction strategies. AGENTREVIEW [13] simulates peer review dynamics with LLM agents, enabling controlled experiments on review outcomes.

2.3 Multi-Agent Systems and Consensus Mechanisms

Recognition that single-model approaches suffer from inherent biases has motivated multi-agent consensus research. D’Arcy et al. [14] introduced MARG, demonstrating that distributing text across specialized agents with internal discussion reduces generic comments from 60% to 29% and doubles high-quality feedback generation.

Guo et al. [15] identify critical collaboration mechanisms: agent profiling, structured communication protocols, and consensus-seeking strategies. Duan et al. [16] propose third-party LLM integration for uncertainty estimation, effectively mitigating hallucinations in multi-agent systems.

Meta-review synthesis represents another development. Traditional peer review employs Area Chairs to synthesize reviewer assessments; computational approaches have begun exploring automated meta-review generation that resolves conflicts across evaluations [17].

2.4 Benchmark Datasets and Evaluation Resources

The seminal PeerRead dataset [18] provides 14.7K paper drafts with accept/reject decisions from ACL, NIPS, and ICLR, alongside 10.7K expert reviews. Ghosal et al. [19] introduced Peer Review Analyze with 1,199 reviews annotated across four dimensions: Section Correspondence, Aspect Category, Functionality, and Significance.

OpenReview [20] provides public access to submissions, reviews, rebuttals, and decisions for ICLR, NeurIPS, and ICML—crucially including rejected submissions essential for understanding acceptance

boundaries. The ORB dataset [21] aggregates over 36,000 submissions and 89,000 reviews with standardized schemas.

For RAG evaluation, RAGBench [22] provides 100K examples with explainable labels. ARES [23] demonstrates that lightweight fine-tuned judges can accurately evaluate RAG systems with minimal human annotations.

2.5 Research Gaps and HorizonBench Positioning

Our analysis identifies five critical gaps that HorizonBench addresses:

Gap 1: Static Evaluation Baselines. Existing systems evaluate against fixed criteria without domain-specific benchmarks. HorizonBench’s Deep Research Agent dynamically constructs contextualized benchmarks by retrieving relevant prior work from Semantic Scholar, arXiv, and OpenReview.

Gap 2: Numerical Score Instability. LLMs produce different scores across repeated runs [10]. HorizonBench adopts grade-based ratings (Strong Accept through Strong Reject) aligned with conference practices, eliminating numerical instability.

Gap 3: Single-Model Bias. Current systems inherit model-specific biases. While MARG [14] uses multiple agents, they employ the same model. HorizonBench implements heterogeneous multi-model consensus with GPT, Claude, and DeepSeek as reviewers, plus Gemini as independent Meta Reviewer.

Gap 4: Process Disconnect. Existing systems collapse multi-stage review into single-pass generation. HorizonBench models the complete pipeline: independent reviews, conflict resolution, and meta-review synthesis.

Gap 5: Limited Rejection Data. Systems predominantly train on accepted papers. HorizonBench integrates OpenReview rejection data into few-shot prompting, calibrating assessment boundaries with examples spanning Strong Accept to Strong Reject.

Additionally, HorizonBench reconceptualizes sparse retrieval results as a feature: papers with little related literature represent potentially high-novelty contributions, triggering upward adjustment in innovation scores rather than assessment failure.

3 System Architecture and Design

HorizonBench is designed to simulate the complete peer review pipeline of top-tier academic conferences. Unlike existing systems that collapse review into single-pass generation, our architecture models the full workflow: from paper understanding through dynamic benchmark construction, multi-dimensional evaluation, to multi-model consensus with meta-review synthesis. This section details the four-phase architecture illustrated in Figure 2.

3.1 Overall Architecture

Table 1 summarizes the four phases of HorizonBench, mapping each to its corresponding real-world peer review component.

Table 1: Four-Phase Architecture Overview

Phase	Name	Core Task	Output
1	Paper Understanding	PDF parsing, content extraction	Paper Schema
2	Deep Research Agent	Dynamic benchmark construction	Contextualized Benchmark
3	Multi-Dimensional Evaluation	4 dimensions, 12 fine-grained metrics	Dimension Ratings
4	Multi-Model Consensus	3 reviewers + meta-review	Final Report

The design philosophy mirrors actual conference review processes: Phase 1 corresponds to a reviewer reading the paper; Phase 2 represents the reviewer’s domain expertise and literature awareness; Phase 3 maps to structured evaluation against review criteria; and Phase 4 simulates the Area Chair synthesizing multiple reviewer opinions into a final decision.

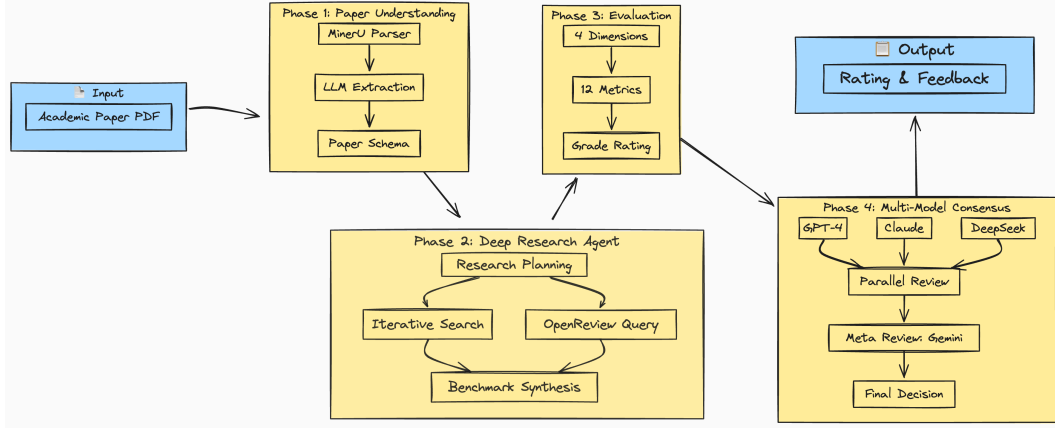


Figure 2: HorizonBench System Overview. The four-phase pipeline transforms raw PDF input into comprehensive evaluation reports through paper understanding, dynamic benchmark construction, multi-dimensional assessment, and multi-model consensus.

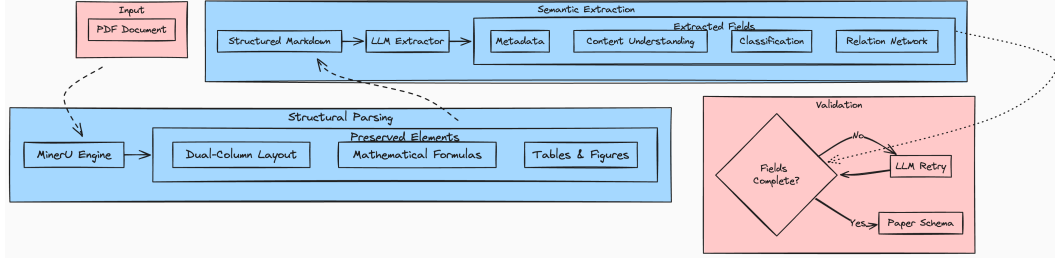


Figure 3: Phase 1: Paper Understanding Pipeline. MinerU handles structural parsing while LLM performs semantic extraction, producing a comprehensive Paper Schema.

3.2 Data Pipeline

3.2.1 Data Source

The primary data source is the ACL Anthology¹, providing academic papers from ACL conferences and workshops. For dynamic benchmark construction, we integrate three external sources: Semantic Scholar API for comprehensive literature search, arXiv for preprints, and OpenReview for real peer review data including both accepted and rejected submissions.

3.2.2 Data Requirements

HorizonBench adopts a few-shot prompting approach, eliminating the need for large-scale training data. Instead, we require: (1) **Input data**: Raw PDF files from the ACL corpus; (2) **Few-shot examples**: Real peer reviews from OpenReview spanning Strong Accept to Strong Reject; (3) **Evaluation data**: A gold-standard subset with expert annotations for system validation, utilizing the PeerRead dataset [18] containing 14.7K papers with 10.7K expert reviews.

3.2.3 Phase 1: Paper Understanding

Phase 1 transforms unstructured PDF documents into structured, machine-readable representations. Figure 3 illustrates this pipeline.

¹<https://aclanthology.org>

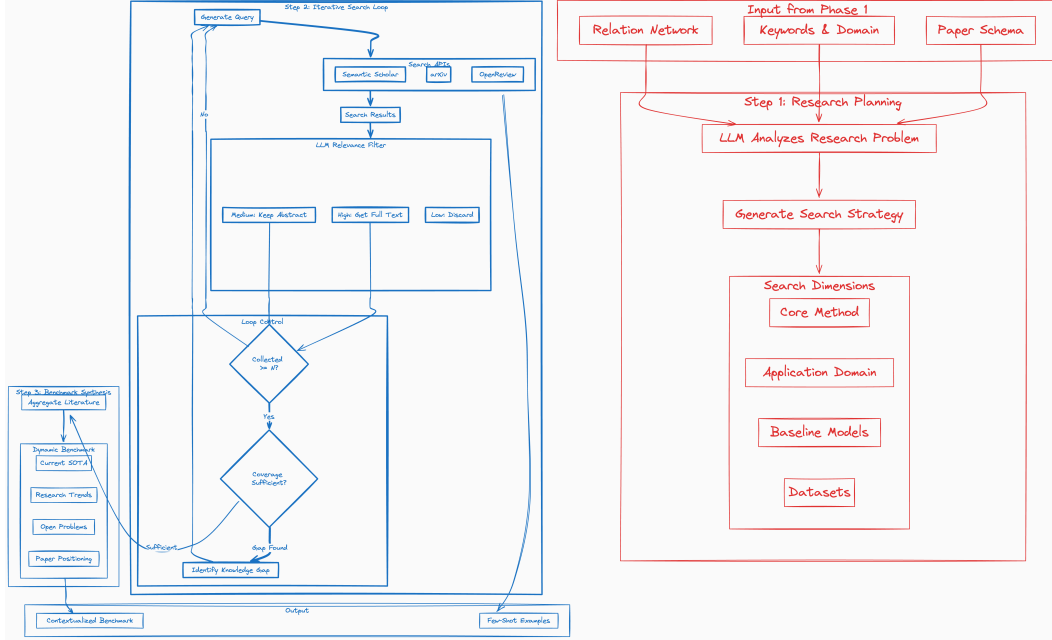


Figure 4: Phase 2: Deep Research Agent. The agent autonomously plans search strategies, iteratively retrieves literature, and synthesizes a contextualized benchmark report.

Technical Selection. We employ MinerU (Magic-PDF) [24] for PDF parsing, selected for its superior handling of academic document structures including dual-column layouts, mathematical formulas, and complex tables. Unlike vision-based OCR approaches that may lose semantic structure, MinerU preserves hierarchical organization through layout-aware parsing.

Processing Pipeline. The workflow proceeds as: PDF → MinerU Parsing → Structured Markdown → LLM Content Extraction → Field Validation → Paper Schema. The LLM extraction step employs GPT-5 or Claude 4.5 Sonnet to identify semantic elements that rule-based parsing cannot capture, such as implicit research contributions or methodology innovations.

Output Schema. The Paper Schema contains four components: (1) `paper_metadata`: title, authors, venue, year; (2) `content_understanding`: abstract, research problem, methodology, contributions, conclusions, limitations; (3) `classification`: domain, sub-domain, paper type, keywords; (4) `relation_network`: compared baselines, used datasets, and seminal references. The `relation_network` specifically supports Phase 2’s “snowball” literature discovery by extracting not just content but relational context.

3.3 Core Models

3.3.1 Phase 2: Deep Research Agent

A critical limitation of existing review systems is static evaluation against fixed criteria without domain-specific context. HorizonBench addresses this through a Deep Research Agent that dynamically constructs benchmarks for each paper. Figure 4 illustrates this agentic workflow.

Agentic Design Philosophy. Inspired by recent advances in research agents [25], we delegate key decisions to the LLM rather than using fixed rules:

- **Query Generation:** LLM dynamically generates multi-round search queries based on evolving understanding
- **Termination Condition:** LLM judges when coverage is sufficient rather than fixed top-K retrieval

- **Relevance Assessment:** LLM determines document relevance and decides retrieval depth (abstract vs. full-text)

Search Infrastructure. The agent queries three APIs: Semantic Scholar (primary, for comprehensive academic search), arXiv (supplementary, for recent preprints), and OpenReview (for real review examples). For highly relevant papers, full-text retrieval is triggered; otherwise, abstracts suffice.

OpenReview Integration. OpenReview provides unique value through access to: reviewer comments, numerical ratings, author rebuttals, and crucially, *rejected submissions*. This rejection data helps calibrate assessment boundaries—systems trained only on accepted papers lack understanding of the acceptance threshold.

Benchmark Synthesis. The agent aggregates retrieved literature into a Dynamic Benchmark Report containing: current SOTA methods, research trends, unsolved problems, and the target paper’s positioning relative to prior work. This contextualizes subsequent evaluation.

Sparse Results as Signal. When the agent finds limited related literature despite extensive search, this signals a potentially novel contribution in an emerging or interdisciplinary area. Rather than penalizing such papers, HorizonBench flags this as a positive indicator for the Novelty dimension.

3.3.2 Phase 4: Multi-Model Consensus

Single-model evaluation inherits model-specific biases and produces inconsistent outputs across runs. HorizonBench implements heterogeneous multi-model consensus, illustrated in Figure 5.

Model Selection and Roles. We employ four distinct model families to maximize diversity:

- **Reviewer 1:** GPT-5 [26] (OpenAI) — state-of-the-art reasoning and multimodal capabilities
- **Reviewer 2:** Claude 4.5 Sonnet [27] (Anthropic) — superior nuanced analysis and instruction following
- **Reviewer 3:** DeepSeek-V3.2 [28] (DeepSeek) — cost-effective with strong academic text understanding
- **Meta Reviewer:** Gemini 3.0 [29] (Google) — fast, independent synthesis avoiding self-evaluation bias

The critical design choice is using a *fourth* model family for meta-review, ensuring the synthesizer has no stake in any individual review.

Parallel Evaluation Strategy. All three reviewers receive identical inputs: the Paper Schema, Dynamic Benchmark, and few-shot examples. Each independently produces dimension ratings and textual feedback. This parallel design prevents anchoring effects where early reviews influence later ones.

Meta-Review Mechanism. The Gemini 3.0-based Meta Reviewer performs four tasks: (1) **Collection:** aggregate all three reviews; (2) **Consensus Identification:** identify points of agreement; (3) **Divergence Resolution:** for disagreements, query the original paper to adjudicate; (4) **Final Decision:** produce overall rating, confidence score, and synthesized feedback.

Confidence Estimation. System confidence derives from three factors: reviewer agreement (3/3 unanimous → High; 2/3 majority → Medium; all different → Low), benchmark coverage (≥ 10 related papers increases confidence), and few-shot match quality (domain-specific OpenReview examples increase confidence).

3.4 Prompt Engineering

3.4.1 Phase 3: Multi-Dimensional Evaluation

Existing systems often conflate distinct quality aspects. HorizonBench enforces orthogonality through a two-layer evaluation structure, shown in Figure 6.

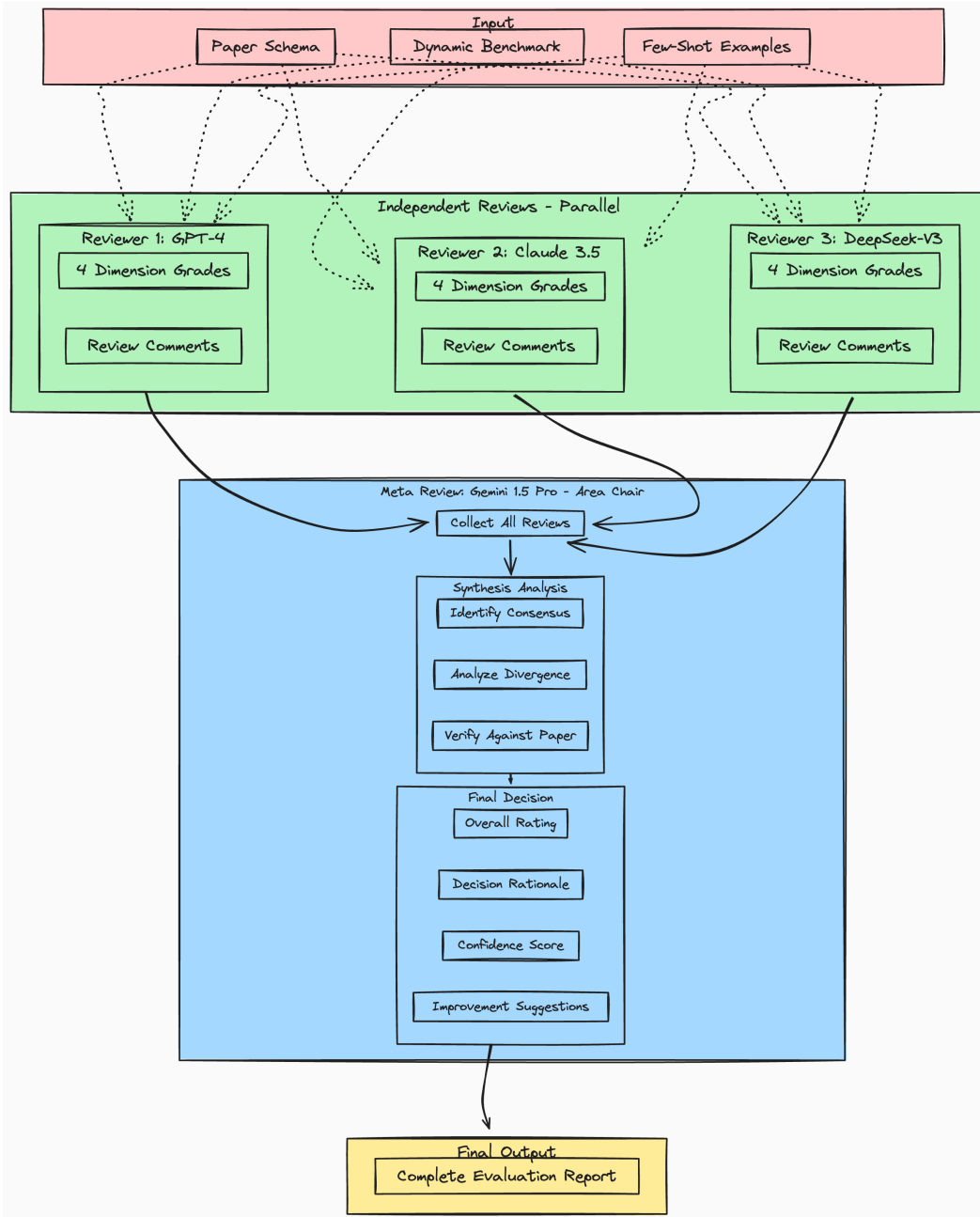


Figure 5: Phase 4: Multi-Model Consensus Mechanism. Three independent reviewers (GPT-5, Claude 4.5 Sonnet, DeepSeek-V3.2) provide parallel assessments, synthesized by a fourth model (Gemini 3.0) serving as Area Chair.

Layer 1: Four Orthogonal Dimensions. Each dimension evaluates one aspect independently:

- **Technical Quality:** Is the methodology correct and rigorous? (Ignores novelty and presentation)
- **Novelty & Contribution:** What is the incremental contribution over SOTA? (Ignores execution quality)
- **Clarity & Presentation:** Is the paper clearly written? (Ignores technical correctness)
- **Soundness & Validity:** Are conclusions adequately supported? (Ignores method complexity)

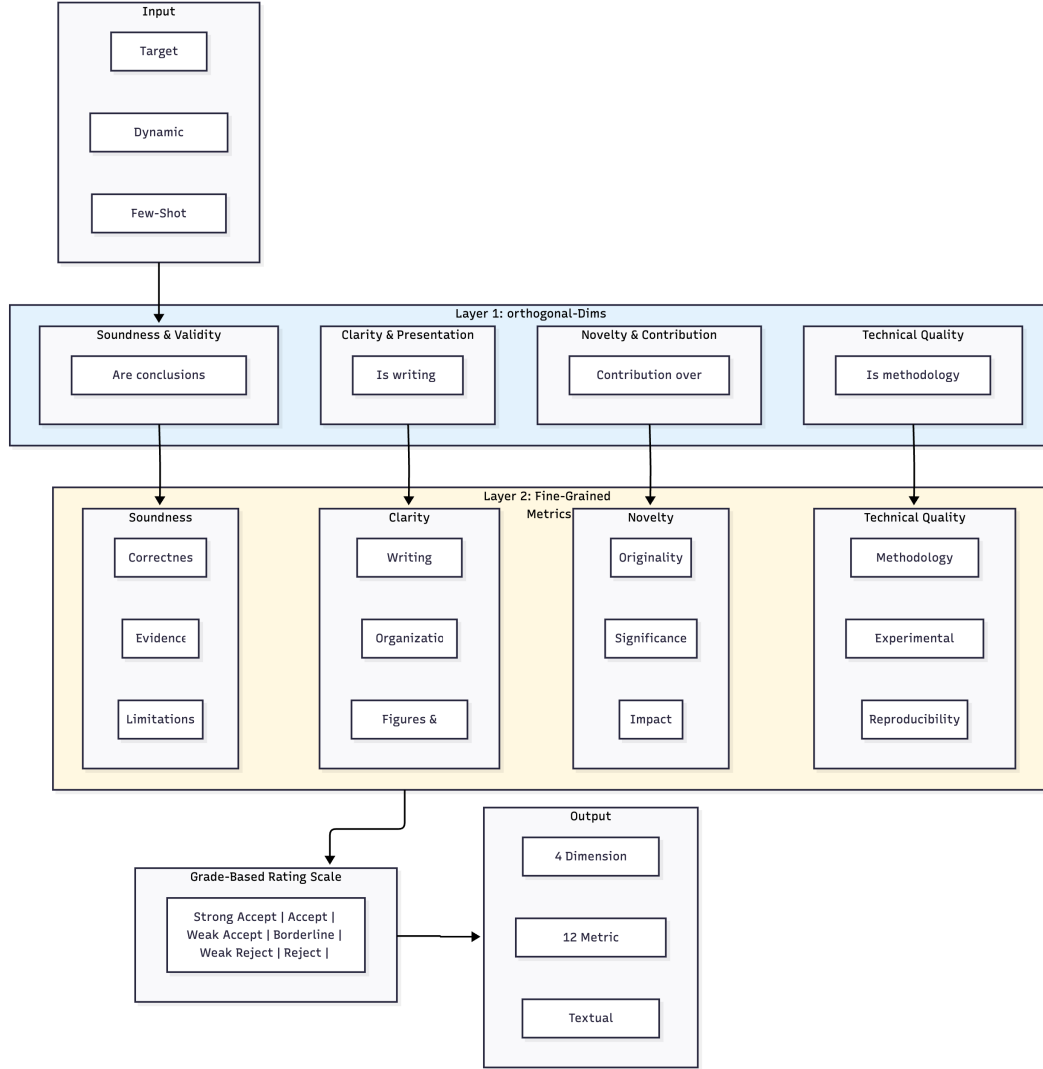


Figure 6: Phase 3: Two-Layer Evaluation Structure. Layer 1 defines four orthogonal dimensions; Layer 2 provides 12 fine-grained metrics. Each dimension is evaluated independently using grade-based ratings.

The orthogonality principle ensures a paper can be rated as “novel but poorly written” or “clear but unsound” without forcing artificial coupling.

Layer 2: Fine-Grained Metrics. Each dimension decomposes into three specific metrics:

Table 2: Two-Layer Evaluation Rubric

Dimension	Fine-Grained Metrics
Technical Quality	Methodology Rigor, Experimental Design, Reproducibility
Novelty & Contribution	Originality, Significance, Potential Impact
Clarity & Presentation	Writing Quality, Organization, Figures & Tables
Soundness & Validity	Correctness of Claims, Evidence Adequacy, Limitations

3.4.2 Rating Rubric

Grade-Based Ratings. A critical design decision is using grades rather than numerical scores. LLMs exhibit numerical instability—repeated runs produce different scores for identical inputs [10]. Grade-based ratings aligned with conference practices provide stability:

Table 3: Seven-Point Rating Scale

Grade	Interpretation	Internal Value
Strong Accept	Significantly exceeds standards, major contribution	10
Accept	Meets top-venue standards, clear contribution	8
Weak Accept	Acceptable with minor issues	6
Borderline	At the threshold, needs improvement	5
Weak Reject	Below standards, notable problems	4
Reject	Does not meet standards, major revision needed	3
Strong Reject	Fundamental flaws, unsuitable for publication	1

Internal numerical values are used only for aggregation and never exposed to the LLM during evaluation.

3.4.3 Prompt Design

Our prompts follow a structured five-component format designed to maximize output consistency and quality.

Component 1: Role Definition. The prompt establishes the LLM as “HorizonBench Reviewer,” an expert academic reviewer with interdisciplinary experience, tasked with objective evaluation following established academic norms.

Component 2: Context Injection. The prompt includes: the Paper Schema from Phase 1, the Dynamic Benchmark from Phase 2, and the four-dimension evaluation rubric. This grounds evaluation in paper-specific context.

Component 3: Few-Shot Examples. We include 2–4 real reviews from OpenReview, strategically sampled:

- 1–2 accepted examples (Strong Accept, Accept) demonstrating positive evaluation
- 1–2 rejected examples showing critical assessment and rejection rationale

This balanced sampling teaches the model to recognize acceptance boundaries.

Component 4: Task Decomposition. Following chain-of-thought principles [30], we decompose evaluation into explicit stages: (1) **Comprehension**: summarize paper’s aim, method, and findings; (2) **Analysis**: assess each dimension with specific evidence; (3) **Synthesis**: prioritize strengths, weaknesses, and actionable recommendations.

Component 5: Output Format. We enforce JSON output schema for structured parsing:

```
{
  "dimensions": {
    "technical_quality": {"grade": "Accept", "justification": "..."},
    "novelty": {"grade": "Strong Accept", "justification": "..."},
    ...
  },
  "overall_rating": "Accept",
  "confidence": "High",
  "strengths": ["...", "..."],
  "weaknesses": ["...", "..."],
  "suggestions": ["...", "..."]
}
```

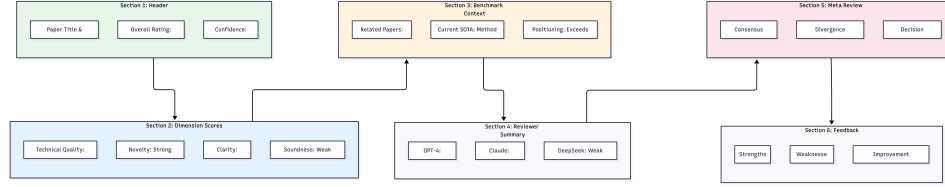


Figure 7: Output Report Structure. The report provides hierarchical information from overall rating through dimension scores, benchmark context, reviewer summary, meta-review synthesis, and actionable feedback.

3.5 Output and User Interface

The final output is a comprehensive evaluation report designed for multiple stakeholders. Figure 7 shows the report structure.

Report Components. The structured report contains six sections:

1. **Header:** Paper metadata, overall rating, and confidence level
2. **Dimension Scores:** Four dimension grades with brief justifications
3. **Benchmark Context:** Number of related papers found, current SOTA, paper positioning
4. **Reviewer Summary:** Individual ratings from GPT-5, Claude 4.5 Sonnet, and DeepSeek-V3.2
5. **Meta Review:** Consensus points, divergence resolution, decision rationale
6. **Feedback:** Itemized strengths, weaknesses, and improvement suggestions

Output Formats. The system supports multiple export formats: JSON for programmatic processing and integration, PDF for archival and formal review records, and interactive web dashboard for exploratory analysis.

Use Cases. The design serves three primary stakeholders: (1) **Conference Organizers:** desk-reject triage and reviewer assignment assistance; (2) **Reviewers:** preliminary analysis reducing cognitive load; (3) **Authors:** pre-submission feedback identifying improvement areas before formal submission.

4 Novelty and Contribution

5 Evaluation Plan

5.1 Evaluation Metrics

5.2 Experimental Setup

6 Ethical Considerations and Limitations

6.1 Algorithmic Bias and Mitigation

6.2 Data Privacy

6.3 Risk of Over-Reliance

6.4 System Limitations

7 Project Timeline

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A Supplementary Material